The mixture form of the Version-4 basket model

Ajit

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Abstract

We replace the Version-4 basket law by its mixture form: each trip draws a taste $z \sim \mathcal{N}(0, I)$ of dimension 16, then a basket from the existing exact law given z. In the current form a taste is weighted by the prior times Z_z, the total weight of all baskets that taste allows. On the Dunnhumby refit this weighting hid a region of 120-product baskets that held most of the probability. Training never visited it and validation mostly missed it: the reported held-out log-likelihood was -38.8 per trip, the value found when the region is included about -53.8. In the mixture form a taste is weighted by the prior alone. Each basket's posterior over tastes is then strongly log-concave, baskets are drawn exactly, and the likelihood estimate is a lower bound with bounded weights. On 64 held-out real trips the mixture form scores -35.83 against -36.78 for the old fit; on synthetic data -27.38 against -27.90.

1 The current form and why it hides modes

A basket S has n_c products in category c and m categories present. The current law is

$$p(S) = \frac{e^{E(S)}}{Z}, \qquad E(S) = E_0(S) + \tfrac{1}{2}\|s(S)\|^2, \tag{1}$$

where E_0 holds the product utilities, within-category terms, size term and competition shift, and $s(S) = \sum_{j \in S} \phi_j - \sum_c (n_c - 1)_+ \psi_c$ is the basket's taste vector (rank 16). The Gaussian identity $e^{\frac{1}{2}\|s\|^2} = \int \mathcal{N}(z; 0, I) e^{z \cdot s}\, dz$ gives

$$p(S) = \int \underbrace{\frac{\mathcal{N}(z; 0, I)\, Z_z}{Z}}_{w(z)} \underbrace{\frac{e^{E_z(S)}}{Z_z}}_{p(S|z)}\, dz, \qquad E_z(S) = E_0(S) + z \cdot s(S). \tag{2}$$

Given z the law $p(S \mid z)$ is exact: Z_z comes from the category-by-size dynamic program and baskets are drawn by backward sampling. The difficulty is the weight $w(z)$.

The tilt. $w(z)$ is the prior multiplied by Z_z. A taste that makes many large baskets attractive has an astronomically large Z_z. On the refit, regions where 120-product baskets dominate had $\log w$ about 50 to 90 nats above the region near the data.

Several peaks. $\log w(z) = \log \mathcal{N}(z; 0, I) + \log Z_z$ is concave plus convex, so it can have several local maxima. A fixed-point search found a giant-basket peak on 63 of 64 held-out trips.

Training cannot see them. The likelihood gradient is

$$\nabla \log p(S) = \nabla E(S) - \mathbb{E}_{w(z)}\big[\nabla \log Z_z\big]. \tag{3}$$

Stochastic maximum likelihood estimated the second term with persistent Gibbs chains started at the observed baskets, about four sweeps per epoch. They never reached the far peak, so its weight was never pushed down.

Validation cannot see them. The held-out estimate of $\log Z$ built its proposal from chains annealed from $\beta = 0.25$. The far chains began in the giant region and left it during annealing. A missed region makes $\hat{Z}$ too small and the reported log-likelihood too high.

64 held-out real trips	production estimator	mode-covering estimate	overstatement
Refit, epoch 1	−38.80	−53.84	15.0
Old fit	−35.92	−36.78	0.86

Table 1: The mode-covering estimate places a Laplace Gaussian at every peak found from eight starts.

2 The mixture form

A trip produces a basket in two steps: draw $z \sim \mathcal{N}(0, I)$, then draw S from $p(S \mid z)$:

$$p(S) = \int \mathcal{N}(z; 0, I)\, p(S \mid z)\, dz, \qquad p(S \mid z) = \frac{e^{E_z(S)}}{Z_z}. \tag{4}$$

The law given z is the one already implemented:

$$E_z(S) = \sum_{j \in S} u_j(z) - \sum_c \Big[\rho_c\tbinom{n_c}{2} + f_c(n_c) + (n_c - 1)_+\big(z \cdot \psi_c - \tfrac{1}{2}\|\psi_c\|^2\big)\Big] - \rho_0(m) + \sum_{c:\, n_c > 0} \iota_c, \tag{5}$$

$$u_j(z) = b_j - \tfrac{1}{2}\|\phi_j\|^2 + z \cdot \phi_j, \tag{6}$$

where b_j carries price, display, mailer, discount, season, history and household taste, and ι_c is the within-category competition shift. For an observed basket, $E_z(S) = E(S) - \frac{1}{2}\|s(S)\|^2 + z \cdot s(S)$.

The form is a mixture: one basket law per taste, weighted by how common that taste is. Products bought together come from shared tastes. There is one component for every z, but all share one formula, so the parameters do not grow with the number of components; the one choice is the dimension of z.

	current form	mixture form
weight of taste z	$\mathcal{N}(z; 0, I)\, Z_z / Z$	$\mathcal{N}(z; 0, I)$
law given z	$p(S \mid z)$, exact	the same
parameters	$b, \phi, \psi, \rho_c, f_c, \rho_0$, price, promotion, competition, history	the same
normaliser to estimate	Z, a 16-dimensional integral	none
drawing a basket	Gibbs chain	exact, two steps

The forms share every parameter, so a current-form fit can start a mixture fit, but they are different laws.

3 Properties

All four follow from one fact: $E_z(S) = a(S) + z \cdot s(S)$ with $a(S)$ free of z.

Proposition 1 (convexity)**.** $\log Z_z$ *is convex in* z*, with* $\nabla_z \log Z_z = \mathbb{E}[s \mid z]$ *and* $\nabla_z^2 \log Z_z = \mathrm{Cov}[s \mid z] \succeq 0$.

Proof. $\log Z_z = \log \sum_S e^{a(S)+z\cdot s(S)}$ is a log-sum-exp of affine functions of z; differentiate twice. □

Proposition 2 (one peak)**.** *For an observed basket S, $\log p(z \mid S)$ is 1-strongly concave. It has a unique maximiser m, which solves $m = s(S) - \mathbb{E}[s \mid m]$, and for every z*

$$\log p(z \mid S) \le \log p(m \mid S) - \tfrac{1}{2}\|z - m\|^2. \tag{7}$$

Proof. $\log p(z \mid S) = -\frac{1}{2}\|z\|^2 + z \cdot s(S) - \log Z_z + \text{const}$, whose Hessian is $-(I + \operatorname{Cov}[s \mid z]) \preceq -I$ by Proposition 1. Strong concavity gives uniqueness and the quadratic bound around the maximiser, where the gradient $s(S) - z - \mathbb{E}[s \mid z]$ vanishes. □

The current form's weight $\log w(z)$ is concave plus convex; the mixture posterior is concave plus linear minus convex, which is concave.

Proposition 3 (exact draws)**.** *Drawing $z \sim \mathcal{N}(0, I)$ and then $S \sim p(S \mid z)$ by backward sampling gives an exact draw from $p(S)$.*

Proposition 4 (lower bound, bounded weights)**.** *With $z_1, \dots, z_K$ drawn from a proposal r covering the posterior,*

$$\hat{p}(S) = \frac{1}{K}\sum_{k=1}^{K} \frac{\mathcal{N}(z_k; 0, I)\, p(S \mid z_k)}{r(z_k)}, \tag{8}$$

$$\mathbb{E}\,\hat{p}(S) = p(S), \qquad \mathbb{E}\log \hat{p}(S) \le \log p(S). \tag{9}$$

If $r \ge \varepsilon\, \mathcal{N}(\cdot\,; m, I)$ with m the maximiser of Proposition 2, then every weight is bounded:

$$\frac{\mathcal{N}(z; 0, I)\, p(S \mid z)}{r(z)} \le \frac{(2\pi)^{d/2}\, p(S)\, p(m \mid S)}{\varepsilon}, \qquad d = \dim z.$$

Proof. Unbiasedness is the importance identity; the inequality is Jensen's. For the bound, $\mathcal{N}(z; 0, I)p(S \mid z) = p(S)\, p(z \mid S) \le p(S)\, p(m \mid S) e^{-\frac{1}{2}\|z-m\|^2}$ by Proposition 2, and $r(z) \ge \varepsilon (2\pi)^{-d/2} e^{-\frac{1}{2}\|z-m\|^2}$. □

So the estimate can understate the fit but on average never overstate it, the opposite of the failure in Section 1. A peak found only approximately loosens the bound but keeps the estimate valid.

4 Training

Training maximises the importance-sampled lower bound plus the production priors, with Adam, and every N steps applies moment matching and whitening. No Markov chain is used.

Objective and gradient. For a batch of B baskets and K draws each ($K = 8$ in the tests),

$$\mathcal{L} = \frac{1}{B}\sum_{t=1}^{B} \log \frac{1}{K}\sum_{k=1}^{K} \frac{\mathcal{N}(z_{tk}; 0, I)\, p(S_t \mid z_{tk})}{r_t(z_{tk})} - \text{priors}, \tag{10}$$

$$\nabla \mathcal{L}_t = \sum_k \bar{w}_{tk}\big(\nabla E_{z_{tk}}(S_t) - \mathbb{E}_{S' \mid z_{tk}}[\nabla E_{z_{tk}}(S')]\big), \tag{11}$$

with normalised weights $\bar{w}$. The expectation is exact: it is the gradient of $\log Z_z$ through the dynamic program.

Per batch.

1. Find each basket's peak m_t by gradient ascent on $\log p(z \mid S_t)$ with backtracking (8 iterations on real data, 15 on synthetic).
2. Estimate $\Sigma_t = (I + \mathrm{Cov}[s \mid m_t])^{-1}$ with Cov from 32 exact draws of $S \mid m_t$.
3. Draw K tastes from $r_t = 0.8\,\mathcal{N}(m_t, 1.2^2\Sigma_t) + 0.2\,\mathcal{N}(m_t, I)$; the second part gives Proposition 4's bound.
4. Compute the K terms $\log p(S_t \mid z_{tk})$ with gradients and take an Adam step.

Moment matching. As in `scripts/fit_interactions.py`: every N steps the product intercepts λ, the size term ρ_0 and the within-category terms ρ_c each take one damped step of their averaged gradient divided by the data rate (product purchase rate, frequency of each category count, mean $\binom{n_c}{2}$), clamped at ± 0.35, ± 0.5, ± 0.5.

Whitening (parameter-expanded EM). At a maximum of the likelihood the tastes pooled over the baskets' posteriors have second moment I. Every N steps we estimate

$$\Sigma = \frac{1}{N_b} \sum_t \sum_k \bar{w}_{tk}\, z_{tk} z_{tk}^\top, \qquad A = \Sigma^{1/2}, \tag{12}$$

over the N_b baskets since the last step, and set $\phi_j \leftarrow A\phi_j$, $\psi_c \leftarrow A\psi_c$, $\lambda_j \leftarrow \lambda_j + \frac{1}{2}(\|A\phi_j\|^2 - \|\phi_j\|^2)$. A model with prior $\mathcal{N}(0, \Sigma)$ equals the rescaled model with prior $\mathcal{N}(0, I)$, and fitting Σ to the pooled posterior is the EM update for the prior's spread, so the step does not lower the likelihood. One term is not compensated: the $\frac{1}{2}\|\psi_c\|^2$ per extra product of a category becomes $\frac{1}{2}\|A\psi_c\|^2$. Adam's state for ϕ and ψ is reset after each rescale.

5 Generation and counterfactuals

Generation. Build b_j from the trip's context, draw $z \sim \mathcal{N}(0, I)$, then $S \mid z$. Repeating gives the trip's basket distribution; averages give incidence, size and category shares with ordinary Monte Carlo error.

Counterfactuals. Change the context (price, display, discount), which changes b_j and ι_c, and redraw S with the same z. Each draw is one shopper with a fixed taste seen before and after the change, a paired comparison.

In the current form a price change also changes Z_z and hence which tastes are common. In the mixture form the taste mix is fixed by the prior and a price change only changes what each taste buys. This is a modelling assumption suited to short-run price and promotion questions; effects on store traffic are outside the model in both forms.

Recommendation. Rank products for a trip by $P(j \in S)$, an average over z of the exact incidence given z. Conditioning on a partly observed basket T uses $p(z \mid T \subseteq S)$. This is not guaranteed log-concave: $\log P(T \subseteq S \mid z)$ is a difference of two convex functions. It needs a peak check.

6 Evidence

All numbers are on the first 64 held-out trips of each world's validation split. The test weeks were not opened.

Exactness. In the test suite's tiny world (6 products, 3 categories, rank 2) and all 11 model variants, $\log p(S \mid z)$ agrees with enumeration of every basket to 10^{-9}, and the importance estimate of $\log p(S)$ (4,000 draws) agrees with enumeration and 48×48 Gauss–Hermite quadrature to 0.007 nats, with effective sample size about 0.91.

Synthetic (48 categories, 522 products, cap 80)	log-lik.	size mean	size 99%	at cap
Data		8.69	22.8	
Current form, fully trained	−27.90			
Mixture, current-form parameters	−29.78	4.32	10	0%
Mixture, 278 steps, no corrections	−27.82	7.00	19	0%
+ 4 epochs moment matching	−27.74	6.93	19.8	0%
+ 8 epochs whitening ($K = 8$)	**−27.38**	7.92	22	0%

Scaling all taste vectors of the last model by c (intercepts compensated) puts the likelihood peak at $c = 1.0$ (-27.42, -27.38, -27.44 at $c = 0.95, 1.0, 1.05$): training has converged on taste strength. Matching the mean size exactly ($c \approx 1.04$) would cost about 0.03 nats per trip.

Real (187 categories, 5,455 products, cap 120)	log-lik.	min. ESS	size mean	size 99%
Data			7.26	42.8
Old fit, current form (mode-covering estimate)	−36.78	0.007		
Old fit's parameters as a mixture	−36.44	0.42	3.81	18
Mixture, 60 steps, no corrections	−35.98	0.35	5.25	26.8
+ 120 steps, moment matching and whitening	**−35.83**	0.28	6.71	35

The last run applied moment matching and whitening every 30 steps. In total the mixture saw 180 batches of 256 trips, about 29% of one epoch (622 batches). Whitening found the pooled tastes 15%, 11%, 9% and 9% too spread at its four rounds, the same convergence as on synthetic data. No generated basket reached the cap.

Cost. One real step on 256 baskets takes 35 to 55 s: about 1.6 s per exact gradient call, 8 to 15 of them in the peak search and 8 in the objective. A damped Newton peak search was not faster (25 s against 30 s) because its curvature came from 16 noisy draws.

7 Limits and next steps

- Shared tastes express products bought together, not cross-category substitution; within a category ρ_c and the competition shift still do. No loss against the current model.
- Generated baskets on real data are 8% too small on average (6.71 against 7.26) and the 99th percentile is 35 against 42.8, after under a third of an epoch.
- The prior is one Gaussian. Real trip types may want a few Gaussian components; Proposition 2 holds within each.
- Whitening leaves one term uncompensated, so it is not an exact EM step.
- With few draws the training objective favours weaker tastes. On synthetic data $K = 16$ reduced the excess spread at whitening (1.11 to 1.07) without changing the held-out result.
- Cost: about 35–55 s per real step against about 6 s for the current step; a full epoch would take 6–9 hours. Warm-starting each basket's peak applies from the second epoch.
- Price and promotion checks must be re-run on the mixture form.

Next: move the code into `v4/` with the enumeration test in `tests/test_law.py`; one full real fit with the mode check and the size-by-period check before any test-week evaluation.